

12 PA 400 MB

SWETON®



12 PA 400 MB is a high-performance 12-inch mid-bass transducer engineered for demanding public address and stage monitoring systems. Built with a 3-inch voice coil, this driver delivers 400W AES power with remarkable control and clarity in the low-mid to midrange frequency spectrum.

With a robust BL factor of 18.15 N/A, Qts of 0.34, and a finely tuned Fs of 58 Hz, it strikes an excellent balance between punchy midbass and dynamic vocal reproduction. The Xmax of 4.15 mm allows sufficient linear excursion, while a large Sd of 510.71 cm² contributes to efficient air movement, making it ideal for venues requiring rich low-end presence.

The transducer maintains clean performance with a high reference efficiency of 2.26%, and SPL around 96 dB, making it a leader in its segment for both clarity and output.

Recommended Enclosure:

Enclosure Type	Volume (Net)	Port Details	Tuning Frequency
Vented	45 – 55 L	4-inch dia x 120 mm length	60 Hz
Sealed	30 – 35 L	N/A	Natural roll-off

HF Driver Pairing Recommendation:

Parameter	Recommendation
Power Handling	60–80W AES
Sensitivity	105–110 dB
Diaphragm	Titanium preferred
Exit Size	1" or 1.4" Compression Driver

Crossover Recommendations

Driver	Type	Frequency (12 dB/oct)
12 PA 400 MB	Low Pass	3.0 – 3.5 kHz
HF Driver	High Pass	1.5 – 1.7 kHz

Applications:

- Portable PA systems
- Stage monitors
- Mid-bass reinforcement in 2-way and 3-way systems
- Indoor and semi-outdoor installations
- School, temple, or auditorium sound setups

Why Choose 12 PA 400 MB?

- Powerful and tight low-mid response suitable for Indian vocals and instruments
- High BL and Qms for excellent transient response
- Designed for rugged environments with stable thermal and mechanical performance
- Well-matched for pairing with popular compression drivers
- Ideal crossover and enclosure compatibility with wide range of cabinets

SPECIFICATIONS & PARAMETERS

Specifications

Nominal Diameter	314 mm
Nominal Impedance	8 Ω
Nominal Power Handling (AES)	400 W
Program Power	800 W
Sensitivity (1W/1m)	96 dB
Frequency Range	55-5500 Hz
Magnet Material	Ferrite
Voice Coil Diameter	76.2 mm (3 in)
Winding Material	COPPER
Former Material	GLASS FIBER
Winding Type	Outside

Mounting Info

Overall Diameter	314 mm
Bolt Circle Diameter	296 mm
Baffle Cutout Diameter	282 mm
Depth	137 mm
Flange and Gasket Thickness	15 mm
Gross Weight	8.5 Kgs

Parameters

Resonant Frequency	Fs	58 Hz
DC Resistance	Re	4.86 Ω
Electrical Q	Qes	0.35
Mechanical Q	Qms	10.96
Total Q	Qts	0.34
Compliance Equivalent Volume	Vas	42.73 Ltrs
Peak Diaphragm Displacement Volume	Vd	0.42 Ltrs
Effective Surface Area of Cone	Sd	510.71 cm ²
Reference Efficiency	η_0	2.27%
Moving Mass including air load	Mms	65.20 gms
Motor Strength	Bl	18.15 T-m
Voice Coil Inductance	Le	0.42 mH
Efficiency Bandwidth Product	EBP	166 Hz
Voice Coil Overhang	Xmax	± 4.15 mm

Recone Kit

Recone Kit Number	REC12PA400MB
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$Z(f, x=0)$ Impedance